



Fungus-mediated biosynthesis and characterization of TiO₂ nanoparticles and their activity against pathogenic bacteria

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ABSTRACT

In the present study, the biosynthesis of TiO₂ nanoparticles (TiO₂ NPs) was achieved by a novel, biodegradable and convenient procedure using *Aspergillus flavus* as a reducing and capping agent. Research on new, simple, rapid, eco-friendly and cheaper methods has been initiated. TiO₂ NPs were characterized by FTIR, XRD, AFM, SEM and TEM studies. The X-ray diffraction showed the presence of increased amount of TiO₂ NPs which can state by the presence of peaks at rutile peaks at 1 0 0, 0 0 2, 1 0 0 and anatase forms at 1 0 1 respectively. SEM observations revealed that synthesized TiO₂ NPs were spherical, oval in shape; individual nanoparticles as well as a few aggregate having the size of 62–74 nm. AFM shows crystallization temperature was seen on the roughness of the surface of TiO₂. The Minimum inhibitory concentration value for the synthesized TiO₂ NPs was found to be 40 µg ml⁻¹ for *Escherichia coli*, which was corresponding to the value of well diffusion test. This is the first report on antimicrobial activity of fungus-mediated synthesized TiO₂ NPs, which was proved to be a good novel antibacterial material.

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1. Introduction

One key aspect of nanotechnology is the development of reliable experimental protocols for the synthesis of nanomaterials over a range of chemical compositions, sizes and high monodispersity. Materials with nano-sized dimensions have attracted considerable attention of the researchers due to their exponential promises in almost all walks of life. Titanium dioxide (TiO₂) nanoparticles are widely used in cosmetics, sunscreen and as a photocatalyst. TiO₂ is technologically very important material especially as dielectrics. Various microbes are known to reduce metal ions to the metals. Minimum time, miniaturization and non-hazardous processes are key parameters for any kind of technology acceptance. Bio-directed syntheses of nanoparticles are of great interest to biologists, chemists and materials scientists to find greener methods of inorganic material synthesis. The formation of extracellular silver nanoparticles by photoautotrophic cyanobacterium, *Plectonema boryanum* had been described [1]. An antibacterial activity test [2] was conducted in order to

confirm the improved bactericidal properties of the composites obtained. Earlier authors investigated the application of TiO₂ in life science [3] and reported that the catalytic and bactericidal properties of TiO₂ can be improved by growing particles of noble metals (Ag, Au or Cu) over its surface [4]. TiO₂ particles are electrical insulators and are difficult to extract from the sprayed surface after treatment. However, their removal can be facilitated by synthesizing composite particles consisting of a magnetic core and a photocatalytic shell.

TiO₂ is stable in aqueous media and is tolerant of both acidic and alkaline solutions. The present method is inexpensive, recyclable, reusable and relatively simple to produce. It can also be synthesized in nanostructure forms more readily than many other catalysts. Biosynthesis approaches that have advantages over conventional methods involving chemical agents associated with environmental toxicity and eco-friendly bio-organisms contain proteins, which act as both reducing and capping agents forming stable and shape-controlled TiO₂ NPs. This method of biological TiO₂ NPs production provides rates of synthesis faster or comparable to those of chemical methods and can potentially be used in various human contacting areas such as cosmetics, foods and sunscreen products applications [5].

Kowshik et al. [6] have identified yeast, *Torulopsis* sp. being capable of intracellular synthesis of PbS crystallite when

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exposed to aqueous Pb^{2+} ions and CdS nanoparticles synthesized intracellularly by using *Schizosaccharomyces pombe*. Bhainsa and D'Souza [7] reported the monodispersed silver nanoparticles within 10 min by using *Aspergillus fumigatus*. The exposure of *Verticillium* sp. fungal biomass to aqueous AgNO_3 solution resulted in the intracellular formation of silver nanoparticles [8], *Fusarium oxysporum* has also been used to synthesize both silver and zirconia nanoparticles [9]. *Lactobacillus* sp. and *Saccharomyces cerevisiae* were used to synthesize TiO_2 NPs [10]. The use of fungi in the synthesis of nanoparticles is a relatively recent addition to the list of microorganisms possessing nanoparticle biosynthesis “ability”. Application of fungi to produce nanoparticles is potentially exciting because of their ability to secrete large amounts of enzymes. The fungus such as, *Aureobasidium pullulans*, *Fusarium* sp. and *F. oxysporum* [11], *Candida guilliermondii* [12], *Chrysosporium tropicum* [13] and yeast *S. cerevisiae* were used for synthesis of silver and gold nanoparticles [14]. The silver nanoparticles produced by the fungus *Amylomyces rouxii* showed antimicrobial activity against *Shigella dysenteriae* type I, *Staphylococcus aureus*, *Citrobacter* sp., *Escherichia coli*, *Pseudomonas aeruginosa*, *Bacillus subtilis*, *Candida albicans* and *F. oxysporum* [15].

The advantages of this method include use of cheap, non-toxic and environmentally benign precursors. Nanoscale titanium dioxide have been used as sunscreens in cosmetics, the primary advantage of using these nanoparticles are well dispersed and transmit visible light, acting as transparent sunblocks. Producing of titanium nanoparticles using microorganisms are preferable compare to physical and chemical methods in aspects of energy, costs and security [16]. In the present investigation, TiO_2 NPs were synthesized using *A. flavus* is a simple aqueous reduction method and characterized using FTIR, XRD, SEM, and AFM. Hence, this process could be suitable for developing a biological process for mass scale production of nanoparticles.

2. Materials and methods

2.1. Synthesis of TiO_2 NPs using *A. flavus*

A. flavus (MTCC no. 7369) culture was obtained from Microbial Type Culture Collection and Gene Bank, Chandigarh, India. All chemicals used were of analytical grade. To prepare biomass for biosynthesis studies the fungus were grown aerobically in a liquid media containing (g/l) KH_2PO_4 , 7.0; K_2HPO_4 , 2.0; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.1; $(\text{NH}_4)_2\text{SO}_4$, 1.0; yeast extract, 0.6; and glucose, 10.0 [7]. The final pH was adjusted to 6.2 ± 0.2 . The flasks were incubated in the mechanical shaker at 200 rpm at 37°C . After 5 days of incubation, the mycelium was separated by filtration and washed thrice with Milli-Q deionized water. The washed mycelium (5 g fresh weight) was challenged with 100 mL of 1 mM TiO_2 (prepared in deionized water) and incubated in shaker at 200 rpm in dark condition at 37°C . Simultaneously, a positive control of incubating the fungus mycelium with deionized water and a negative control containing only TiO_2 solution were maintained under same conditions [10].

2.2. Instruments used

Characterization involved FTIR analysis of the dried powder of synthesized TiO_2 NPs by scanning in the range $350\text{--}4000\text{ cm}^{-1}$ at a resolution of 4 cm^{-1} . These measurements were carried out on a Perkin-Elmer Spectrum One instrument in the diffuse reflectance mode at a resolution of 4 cm^{-1} in KBr pellets. For comparison, a drop of 20% fungal synthesized TiO_2 nanopowder was mixed with KBr powder and pelletized after drying properly. The fungal mycelium embedded with TiO_2 NPs was freeze-dried, powdered and used for XRD analysis. The spectra were recorded in Philips® automatic

X-ray Diffractometer with Philips® PW 1830 X-ray generator. The diffracted intensities were recorded from 30° to 80° 2θ angles. Topography was studied using an AFM (Veeco PicoForce) working in the contact mode. AFM images have been processed using WSxM software ver. 4.0 [17]. AFM was used to characterize the uniformity and grain size of TiO_2 films deposited on different substrates. The freeze-dried mycelia mats (TiO_2 NPs synthesized sample) were mounted on specimen stubs with double-sided adhesive tape and coated with gold/palladium in a sputter coater and examined under Philips® XL 30 SEM at 12–15 kV with a tilt angle of 45° . The size of the nanoparticles is confirmed by using TEM analysis (transmission electron microscopy – Hitachi H-7100 using an accelerating voltage of 120 kV and methanol as solvent).

2.3. Antimicrobial activity

2.3.1. Agar diffusion assay

The antimicrobial activity of synthesized TiO_2 NPs was evaluated against *S. aureus* (MTCC-3160), *E. coli* (MTCC-1721), *P. aeruginosa* (MTCC-1034), *Klebsiella pneumoniae* (MTCC-4030) and *B. subtilis* (MTCC-1427). Exactly 0.2 ml of fresh cultures of each organism was inoculated into 5 ml of sterile nutrient broth (Hi Media) and incubated for 3–5 h to standardize the culture to McFarland standards (10^6 CFC/ml). Three replicates of respective microorganism were prepared by spreading 100 μl of revived culture on MHA (Mueller Hinton Agar-Hi Media) media with the help of spreader. Well was made having a diameter of about 7 mm and 50 μl samples of synthesized TiO_2 NPs were added in one well and 50 μl tetracycline (2000 units) was also added in a separate well as standard. The petri plates were incubated at 37°C for 24 h in incubator during which activity was evidenced by the presence of a zone of inhibition (mm) surrounding the well.

2.3.2. Minimum inhibitory concentration (MIC)

The microdilution method for estimation of MIC values was carried out to evaluate the antimicrobial activity. The MIC values were determined on 96-well microdilution plates and according to published protocols [18].

2.4. Statistical analysis

All experiments were carried out in triplicate and representative data is presented in this study. For the experiments on antimicrobial activity, arithmetic mean values were considered for data analysis. For comparison of the data obtained by the two types of nanoparticles, the unpaired ‘t’ test was performed. All the statistical analysis was done by PASW Statistics 18 Release Version 18.0.0, 2009.

3. Results and discussion

3.1. Fourier transformed infrared (FTIR) spectroscopy analysis

FTIR studies of the TiO_2 NPs showed the characteristics of the formation of high purity product. The FTIR spectra (Fig. 1) of these nanoparticles showed the peaks only corresponding to TiO_2 . The peak observed at 590 cm^{-1} is due to the vibration of the Ti–O–O bond. The FTIR spectrum firmly suggests the presence of Ti–O bonds, and the absence of peroxy, and –OH groups in the final product. The TiO_2 NPs prepared by this method are of good quality and can be used for the further applications. A broad intense band at $\sim 3430\text{ cm}^{-1}$ in the spectra can be assigned to the N–H stretching frequency arising from the peptide linkages present in the proteins of the biosynthesis of *A. flavus* using TiO_2 . A strong absorption peak at 1779 cm^{-1} is characteristic of asymmetrical C=O coupled (str.)

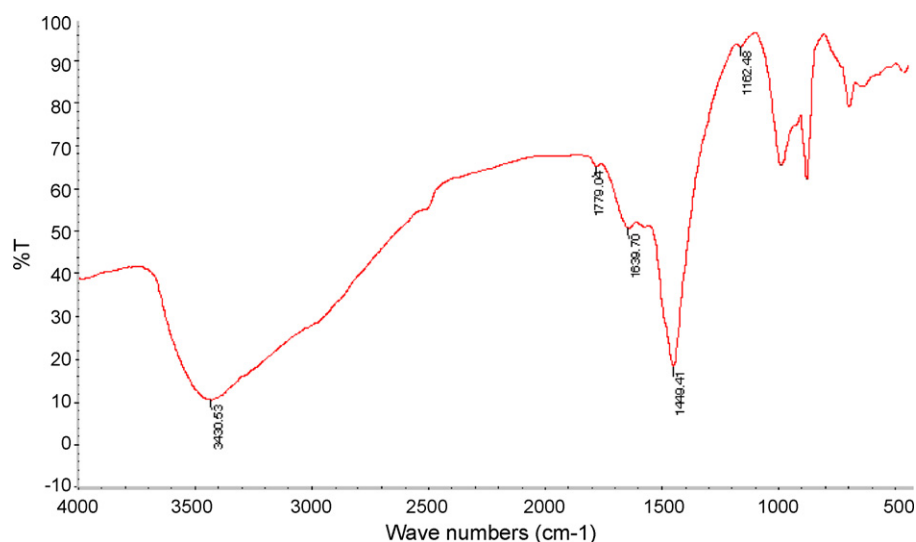


Fig. 1. FTIR peaks of TiO_2 nanoparticles.

vibration of anhydride group. FTIR was used to observe a change in the carbonyl region ($1600\text{--}1800\text{ cm}^{-1}$) and the appearance of characteristic bands at 1779 cm^{-1} and 1639 cm^{-1} assignable to amide I and amide II, as well as a strong decrease in the relative intensity of the bands (characteristic of the symmetric $\text{C}=\text{O}$ stretching vibrations from maleic anhydride groups). New bands at 1779 and 1639 cm^{-1} characteristic for the amide I and amide II proves the presence of amino acid residues from fungal matrix of *A. flavus* (Fig. 1).

3.2. X-ray diffraction (XRD) studies

XRD studies indicate that the materials synthesized were pure anatase TiO_2 phase and the crystal structures agree well with the corresponding reported JCPDS data (JCPDS powder diffraction data card no. 81-84). Line broadening of the diffraction peaks is an indication that the synthesized materials are in nanometer range. To determine the crystal phase composition of the titanium particles formed from the TiO_2 , X-ray diffraction (XRD) measurements were carried out over the diffraction angle (2θ) $10^\circ\text{--}80^\circ$ (Fig. 2). The pattern of the sample showed the presence of peaks ($2\theta = 30.428^\circ$, 35.507° , 38.250° , and 40.169°), which is regarded as an attributive indicator of biologically synthesized TiO_2 NPs crystallites. The presence of TiO_2 results in the formation of Bragg peaks at 30.428° , 35.507° , 38.250° , and 40.169° corresponding to the presence of peaks at rutile peaks at 100, 002, 100 and anatase forms at

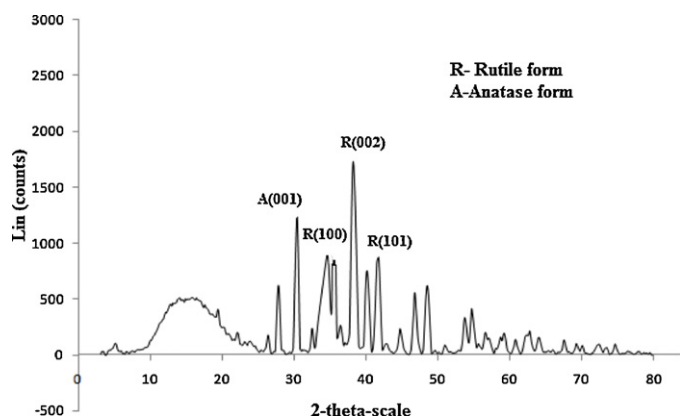


Fig. 2. XRD patterns for the synthesized TiO_2 NPs from *A. flavus*.

101 respectively. The mean particle diameter of the nanoparticles was calculated from the XRD pattern using the Scherrer equation: $D = K\lambda / \beta_{1/2} \cos \theta$, K is the shape constant, λ is the wavelength of the X-ray, $\beta_{1/2}$ and θ are the half width of the peak and half of the Bragg angle, respectively [19].

The crystallite sizes were calculated from Scherrer formula applied to the major intense peaks and found to be in the range of

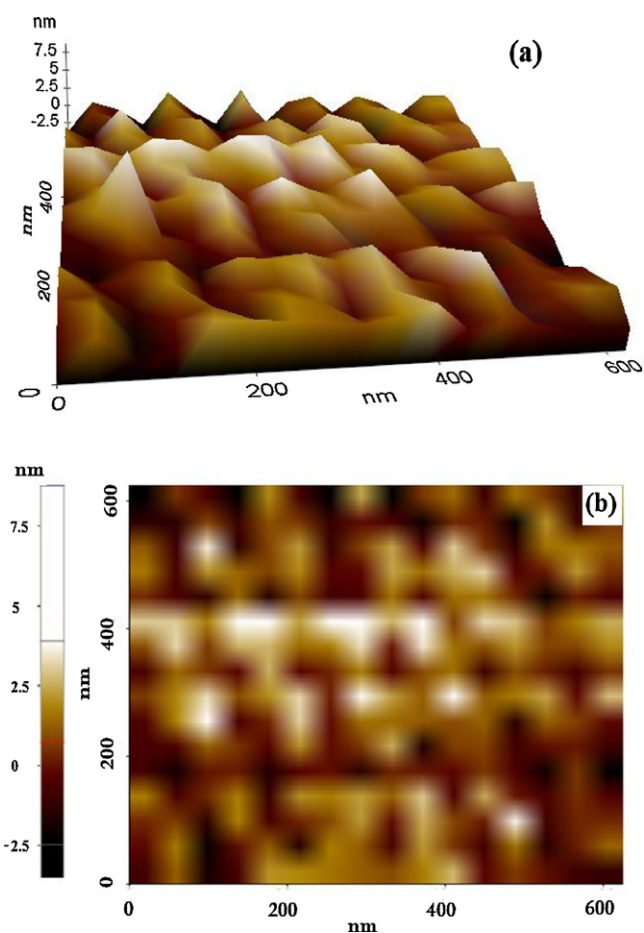


Fig. 3. AFM images of (a) topography of the surface of synthesized TiO_2 NPs at cross sectional view, and (b) top view of the synthesized TiO_2 NPs particles.

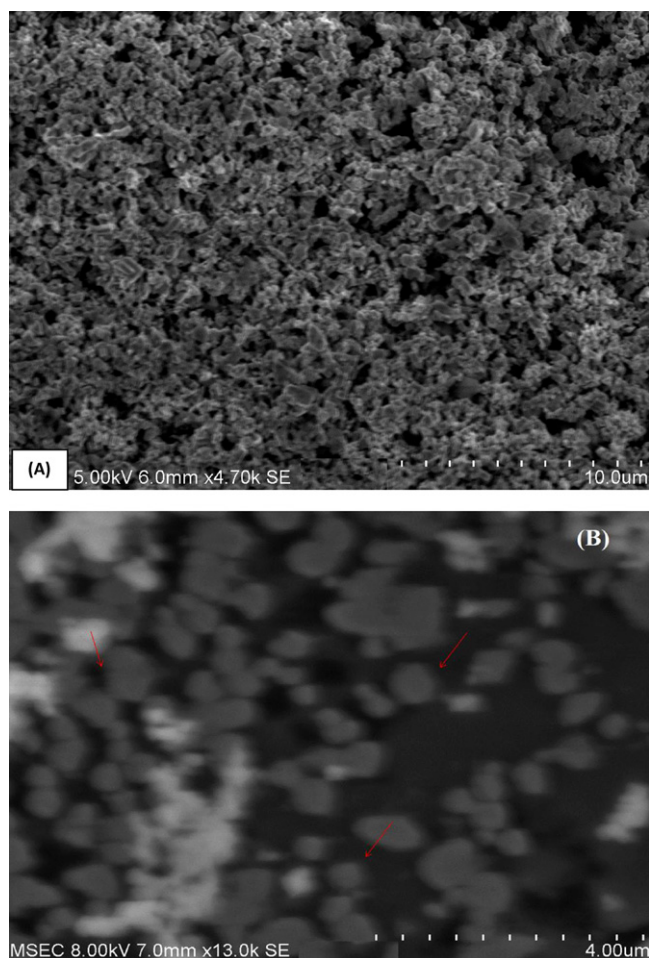


Fig. 4. SEM images of synthesized TiO₂ NPs at room temperature (A) TiO₂ NPs sample at 8000 $\times$, and (B) TiO₂ NPs sample at 13,000 $\times$.

62–74 nm. The lattice parameters calculated are also in accordance with the reported value.

3.3. Atomic force microscopy (AFM)

A small volume of sample was spread on a well-cleaned glass cover slip surface mounted on the AFM stub, and was dried with nitrogen flow at room temperature. Images were obtained in tapping mode using a silicon probe cantilever of 125 μ m length, resonance frequency 209–286 kHz [20]. Porosity, roughness and fractal dimension were evaluated by analyzing the AFM images. The surface TiO₂ NPs were analyzed and the aggregated structures having considerable surface roughness. The non-contact AFM provides suitable data regarding the biological preparation and the application of *A. flavus* in the synthesis of the metal based nanoparticles shows the heights of particles are about 10 nm. The AFM images were also used for the analysis of the fractal behavior as deposited and annealed films. Porosity, roughness and fractal dimension were evaluated by analyzing the AFM images using post image processing software [21]. The surface TiO₂ NPs except that of the as-deposited one consisted of aggregated structures having considerable surface roughness (Fig. 3). The surface area of the synthesized TiO₂ NPs increased dramatically due to the enzymes present in the *A. flavus*, which is indicated by the redox enzymes present in the organisms. In accordance with the above results, the same influence of TiO₂ concentration and crystallization temperature was seen on the roughness of the surface of TiO₂. By means of

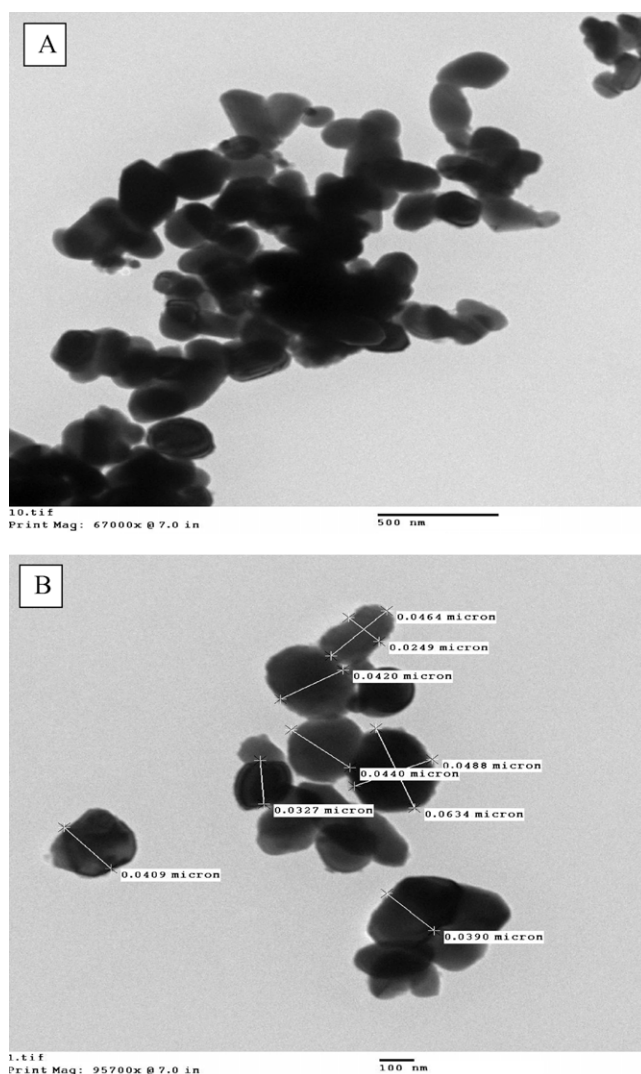


Fig. 5. TEM images of synthesized TiO₂ NPs using *A. flavus* (A) TiO₂ sample at 67,000 $\times$, and (B) TiO₂ NPs sample at 95,700 $\times$.

AFM no linear trend in roughness was observed, but it is proved that the highest TiO₂ concentration result in the formation of smoother layers [22].

3.4. Scanning electron microscope (SEM)

The surface morphology of TiO₂ nanoparticles was studied using SEM. The nanoparticles were distributed uniformly on the surface with formation of aggregated nanoparticles. It shows that the nanoparticles were densely dispersed with a narrow range of dispersion. Particles were of size with smooth and rough surface (Fig. 4). The nanoparticles were in the structure of a spherical and its average particle has dimensions size approximately from 200 to 2000 nm. The observed micrograph shows synthesized TiO₂ NPs aggregates and spherical nanoparticles in the average size range 62–74 nm. The nanoparticles were not in direct contact even within the aggregates, indicating stabilization of the nanoparticles. The result indicates the reduction process being held in the surface. The mycelia, matted together, were more immobile, and more capable of binding TiO₂ NPs than that of the external cellular substances that distributed in the inter-mycelial space.

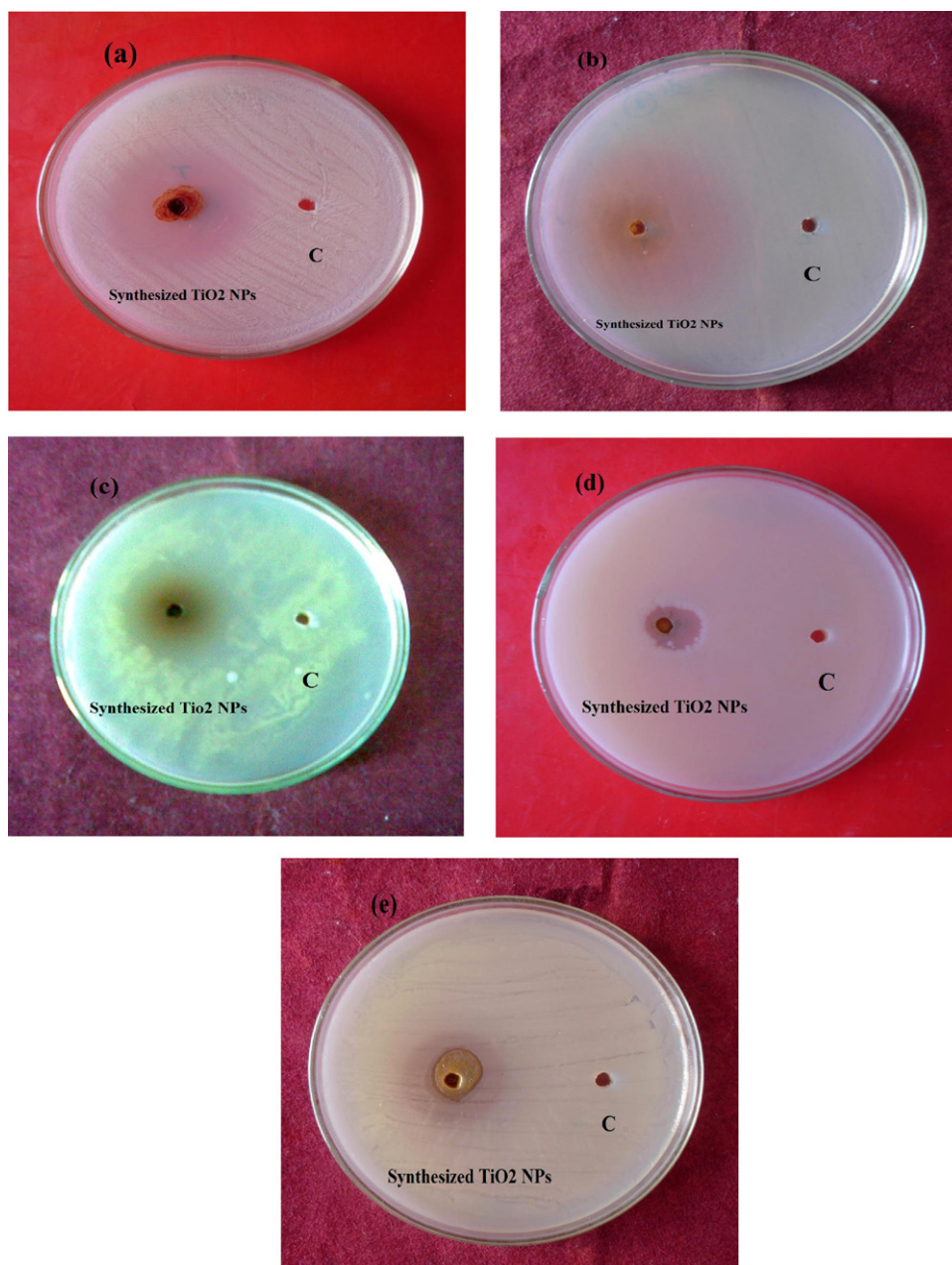


Fig. 6. Zone of inhibition of TiO₂ NPs against (a) *Staphylococcus aureus*, (b) *Escherichia coli*, (c) *Pseudomonas aeruginosa*, (d) *Klebsiella pneumoniae*, and (e) *Bacillus subtilis*.

3.5. Transmission electron microscope (TEM)

The TEM images (Fig. 5A) showed polydisperse nanoparticles with different shapes such as spherical and hexagonal. The particles are distributed in the size 60 ± 5 nm range (Fig. 5B). The titanium nanoparticles being formed using *Lactobacillus* strain and the TEM micrograph clearly illustrates individual nanoparticles were found almost spherical in shape having a size of 40–60 nm [23].

3.6. Agar diffusion and minimum inhibitory concentration (MIC)

The aim of this study was to evaluate the antimicrobial activity of synthesized TiO₂ NPs by *A. flavus* and their dependency of that activity on selected bacterial species, *S. aureus* (25 mm), *E. coli* (35 mm), *P. aeruginosa* (27 mm), *K. pneumoniae* (18 mm) and *B. subtilis* (22 mm) (Table 1 and Fig. 6a–e). The standard antibiotic tetracycline was used as a control. The synthesized TiO₂ NPs was

tested for antimicrobial and the results were compared with control. Each test was performed in triplicates. The well diffusion test was carried out in order to know the zone of inhibition of the respective bacterial species and zones were graphed. The clinical pathogens were tabulated for the antibiogram, showing the zone

Table 1

Zone of inhibition (mm) and MIC ($\mu\text{g ml}^{-1}$) of *A. flavus* synthesized TiO₂ nanoparticles against various microorganisms.

Microorganism	Strain no.	<i>A. flavus</i> synthesized TiO ₂	
		Zone of inhibition (mm)	MIC ($\mu\text{g ml}^{-1}$)
<i>S. aureus</i>	MTCC-3160	25	40
<i>E. coli</i>	MTCC-1721	35	40
<i>P. aeruginosa</i>	MTCC-1034	27	80
<i>K. pneumoniae</i>	MTCC-4030	18	70
<i>B. subtilis</i>	MTCC-1427	22	45

of inhibitions (ZOI) of Gram positive bacteria larger than that of the Gram negative bacteria. A further objective was to gain knowledge about synthesized TiO₂ NPs to appraise a possible application of this material as antimicrobial agents. Antibacterial test were performed using the well diffusion test [24] and MIC [25]. Recently, nanoparticles have found applications in antibacterial effects. It has been shown that extracellularly produced silver or gold nanoparticles using *F. oxysporum*, can be incorporated in several kinds of materials such as cloths. These cloths with silver nanoparticles are sterile and can be useful in hospitals to prevent or to minimize infection with pathogenic bacteria such as *S. aureus* [26]. TiO₂ also showed antibacterial activity against *E. coli* [27–30] and *Bacillus megaterium* using environmental light [4]. The well diffusion test showed the zone of inhibition in the Gram positive and as well as Gram negative. The MIC observed in the present study for synthesized TiO₂ NPs were 40 µg ml⁻¹ for *S. aureus* (MTCC-3160), 40 µg ml⁻¹ for *E. coli* (MTCC-1721), 80 µg ml⁻¹ for *P. aeruginosa* (MTCC-1034), 70 µg ml⁻¹ for *K. pneumoniae* (MTCC-4030) and 45 µg ml⁻¹ for *B. subtilis* (MTCC-1427) (Table 1). Bacteria-killing experiments indicate a significantly higher proportion of all tested pathogens including *S. aureus*, *Shigella flexneri* and *Acinetobacter baumannii* were eliminated by the new visible-light-activated TiO₂ NPs with higher bacterial interaction property [31]. The bactericidal effect of TiO₂ generally has been attributed to the decomposition of bacterial outer membranes by reactive oxygen species (ROS), primarily hydroxyl radicals (•OH), which leads to phospholipid peroxidation and ultimately cell death [32,33]. It was proposed that nanomaterials that can physically attach to a cell can be bactericidal if they come into contact with this cell [34]. If the membrane of a bacterium is compromised, the cell may repair itself or, if the scratch is severe, the cell component may release and eventually the cell will die [35]. The impact of nanomaterials on living cells, including bacteria, can also be elucidated by the interactions between the nanomaterial and the individual cell components. The first interaction between a material and a cell is at the membrane interface; some nanoparticles were suggested to embed themselves in the cell membrane [36]. The antibacterial activity of fungus *Phytophthora infestans* synthesized silver nanoparticles showed zone of inhibition against the clinically isolated human pathogenic bacteria like *S. dysenteriae* (20 mm), *E. coli* (17 mm), *Salmonella typhi* (19 mm), *K. pneumoniae* (20 mm), *Proteus vulgaris* (16 mm), *B. subtilis* (18 mm) and *S. aureus* (14 mm) and MIC which varied from 0.157 to 0.625 µg ml⁻¹ [37]. The extracellular synthesis of silver nanoparticles by *Rhizopus stolonifer* and antibacterial activity against multidrug resistant strains *P. aeruginosa* (P1 and P2) showed zone of inhibition (mm) of about 33 mm and 30.5 mm in diameter [38]. Mritunjai et al. [39] reported that antibacterial effect was size and dose dependent and more pronounced activity against Gram negative bacteria than Gram positive bacteria.

Gogniat et al. [40] pointed out that the aggregation of bacteria onto TiO₂ particles was the key step in photo killing because hydroxyl radicals have an extremely short lifetime (10⁻⁹ s) and must be generated near the cell membrane. The membrane bound (as well as cytosolic) oxidoreductases and quinones might have played an important role in the process of synthesis [41]. Along with this, a number of simple hydroxy/methoxy derivatives of benzoquinones and toluquinones are elaborated by lower fungi (especially *Penicillium* and *Aspergillus* species). *Aspergillus* sp. might be treasuring any other such quinone because it belongs to the same class of fungi thereby facilitating the redox reactions due to its tautomerization. The transformation seems to be negotiated at two distinct levels, at the cell membrane level immediately after addition of the TiO(OH)₂ solution which triggers tautomerization of quinones and low pH sensitive oxidases and makes molecular oxygen available for the transformation. Once entered into the cytosol, the TiO(OH)₂ might have triggered the family of oxygenases

harboured in the endoplasmic reticulum (ER), chiefly meant for cellular level detoxification through the process of oxidation/oxygenation [10,42]. Hence the biosynthesis of TiO₂ NPs was achieved by *A. flavus* as a reducing and capping agent.

4. Conclusion

A simple and inexpensive technique has been established to prepare nanocrystalline TiO₂ NPs powder using *A. flavus* at room temperature. The synthesized TiO₂ NPs were characterized by XRD, FTIR, SEM and AFM studies, which concluded the formation of TiO₂ NPs. The antibacterial tests were performed and MIC values of the test bacterial species were tabulated. This technique will help in synthesis of other metal oxide by the said procedure in future.

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